

AI, Workforce Change, and Student Pathways

A Research Synthesis for K–16 Counselors

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Abstract

Artificial intelligence is reshaping the United States labor market, but recent evidence drawn from major employer surveys, intergovernmental research, peer-reviewed field experiments, and federal occupational data indicates that AI is changing tasks and skill requirements far faster than it is producing widespread, immediate job displacement. Workers with AI fluency earn measurable wage premiums, employer demand for skills evidence is rising even as some degree requirements loosen, and human capabilities such as communication, judgment, and ethics remain central to AI-exposed work. For California’s K–16 counselors, this evidence base supports a renewed advising frame: helping students develop AI literacy as a baseline employability skill while emphasizing applied experience through dual enrollment, Career Technical Education, College and Career Access Pathways, and college promise programs. This paper synthesizes recent research from international and domestic sources, identifies pathway infrastructure available across California, and offers considerations for counseling practice grounded in current evidence.

Keywords: *artificial intelligence, K–16 counseling, community college, workforce development, dual enrollment, AI literacy, career education*

Introduction

The integration of artificial intelligence into workplace operations has moved beyond speculation into measurable change. McKinsey & Company (2024, 2025) report that 78% of organizations used AI in at least one business function in 2024, and that 92% of organizations plan to increase AI investment—despite only 1% characterizing themselves as mature in AI deployment. For students entering the workforce over the next decade, AI is not an exotic specialty; it is becoming a routine feature of work across nearly every sector.

Yet the popular framing of AI as a job-replacement story does not match the strongest evidence available. Survey data from the World Economic Forum (2025) and LinkedIn Economic Graph (2025) consistently indicate that change is concentrated in skills and tasks rather than in net headcount. The Organisation for Economic Co-operation and Development (2024, 2025) finds that even highly AI-exposed roles continue to demand significant human and managerial capabilities. Wage and credential signals from PwC (2025) suggest that the labor market is rewarding workers who can use AI productively and demonstrate evidence of capability rather than those who simply possess legacy credentials.

For counselors working in K–16 settings, these findings carry direct practice implications. Advising conversations grounded in fear of replacement misalign with the evidence; conversations grounded in task analysis, skill development, and applied experience align with it. This paper synthesizes the most relevant recent research, with attention to California’s pathway infrastructure, and concludes with considerations for counseling practice. Several limitations of the current evidence base are also noted, since the most reliable evidence today describes adoption, exposure, skills demand, and short-run productivity—not a final ledger of jobs lost versus jobs created.

Workforce Evidence

Tasks Are Changing Faster Than Jobs

The strongest cross-source finding in recent labor research is that AI is reshaping tasks and skill requirements faster than it is producing immediate, widespread job displacement. The World Economic Forum (2025) projects that 39% of workers’ core skills will change by 2030, with 59 of every 100 workers expected to need training during that period. LinkedIn’s Work Change Report (2025) corroborates this trajectory: 70% of skills used in most jobs are projected to change by 2030, and AI literacy skills added by LinkedIn members rose 177% since 2023.

PwC’s Global AI Jobs Barometer (2025) extends the analysis by comparing high- and low-exposure occupations. In the most AI-exposed roles, the skills sought by employers are changing 66% faster than in the least-exposed occupations. Workers with documented AI skills earn an average 56% wage premium, and degree requirements have fallen faster in AI-exposed jobs between 2019 and 2024 than in other occupations. These signals indicate not a collapse of work but a rapid reweighting of what employers value.

Field experimental evidence reinforces this view. Brynjolfsson, Li, and Raymond (2023), in a National Bureau of Economic Research working paper, found that generative AI raised productivity by 14% on average and by 34% among novice and lower-skilled workers in their study, suggesting that AI can serve as an effective scaffold for early-career workers. A subsequent study by Dell’Acqua and colleagues (2025), published in *Organization Science*, found that within AI’s task-capability frontier, participants completed 12.2% more tasks 25.1% faster; outside that frontier, however, participants were 19% less likely to be correct. The implication for counselors is that AI tools require judgment and verification habits, not simply tool access.

Table 1

Recent evidence on AI workforce change, with relevance for K–16 counseling practice

Evidence point	Source	Key statistic	Relevance for counselors
Employers expect major skill disruption this decade	World Economic Forum (2025), <i>Future of Jobs Report</i>	39% of workers’ core skills are expected to change by 2030; 59 of 100 workers expected to need training	Advising should focus on reskilling readiness, not one-time career choice

Evidence point	Source	Key statistic	Relevance for counselors
Skills are changing quickly across occupations	LinkedIn (2025), <i>Work Change Report</i>	70% of skills used in most jobs projected to change by 2030; AI literacy skills added by members rose 177% since 2023	Students should expect ongoing skill refresh in nearly every field
AI-exposed jobs are experiencing faster skill churn	PwC (2025), <i>Global AI Jobs Barometer</i>	Skills sought by employers are changing 66% faster in the most AI-exposed occupations than in the least-exposed	Curriculum and advising cycles need to move faster
AI skills carry a measurable labor-market premium	PwC (2025), <i>Global AI Jobs Barometer</i>	Workers with AI skills earn an average 56% wage premium; degree requirements fell faster in AI-exposed jobs (2019–2024)	Counselors should emphasize demonstrable skills, not degree title alone
Adoption is moving faster than organizational readiness	McKinsey & Company (2024, 2025)	78% of organizations reported AI use in 2024; 71% reported regular gen-AI use; 92% plan to increase AI investment, but only 1% describe themselves as mature	Students need to be prepared for imperfect, uneven, real-world implementation
Highly AI-exposed jobs still demand human and managerial capabilities	OECD (2024)	In high-AI-exposure vacancies, 72% demand at least one management skill; 67% one business-process skill; over 50% social, emotional, and digital skills	Human skills remain central even in AI-heavy work
Early business evidence suggests task change more than immediate staffing cuts	OECD (2025), SME survey	83% of SMEs reported no effect on total staff need; 19.7% reported increased skill needs (largest increases in data analysis and creativity)	Students should prepare for role redesign rather than only job-replacement narratives
Younger workers see AI skills as necessary, but not sufficient	Deloitte (2025)	59% of Gen Z and 62% of millennials say gen-AI skills are required for advancement; 86% and 85% respectively say soft skills are	Advising should pair AI fluency with leadership, empathy, networking, and communication

Evidence point	Source	Key statistic	Relevance for counselors
		required	
Employers report that their skills systems are not keeping up	Gartner (2025)	48% of HR leaders say demand for new skills is evolving faster than existing talent structures can support; “hiring for promise” outperforms “hiring for proficiency” by 1.9x	Growth mindset and learning agility matter more in advising conversations
AI can boost productivity, especially for newer workers	Brynjolfsson, Li, & Raymond (2023), NBER	Generative AI raised productivity by 14% on average and 34% for novice and lower-skilled workers	AI can function as a scaffold for learners and early-career workers
AI help is powerful but uneven	Dell’Acqua et al. (2025), <i>Organization Science</i>	Within AI’s capability frontier, participants completed 12.2% more tasks 25.1% faster; outside the frontier, 19% less likely to be correct	Students need judgment, verification habits, and domain knowledge—not just tool access
Adjacent digital and analytical roles continue to grow; some routine coding roles contract	U.S. Bureau of Labor Statistics (2025), <i>Occupational Outlook Handbook</i>	Data scientists projected to grow 34% (2024–2034); information security analysts 29%; computer programmers projected to decline 6%; total U.S. employment projected to grow by 5.2 million jobs	Students should be guided toward AI-adjacent growth fields rather than only generic coding roles

Adoption Outpaces Organizational Readiness

Adoption metrics confirm that AI is now a mainstream feature of work, but organizational capacity to integrate it lags. McKinsey & Company (2024) reports that 71% of organizations regularly use generative AI in at least one business function, while McKinsey (2025) finds that 92% of companies plan to increase AI investment yet only 1% describe themselves as mature in deployment. The Organisation for Economic Co-operation and Development (2025), in a survey of small and medium enterprises, found that 83% of small and medium enterprises reported no effect on total staff need following AI adoption, while 19.7% reported increased skill needs—particularly in data analysis and creativity.

These findings together suggest that students entering the workforce will encounter imperfect, uneven, and sometimes immature AI implementations across employers. Preparation for that reality involves not

only AI fluency but also the capacity to work productively in environments where AI tools are inconsistently applied.

Human Skills Remain Central

Even in highly AI-exposed roles, employer demand for human skills remains robust. The OECD (2024) reports that in high-AI-exposure vacancies, 72% demand at least one management skill, 67% require at least one business-process skill, and over 50% specify social, emotional, or digital competencies. Deloitte's 2025 Gen Z and millennial survey shows that 86% of Gen Z and 85% of millennials say soft skills are required for advancement—closely paralleling rates for AI skills (59% and 62%, respectively). Gartner (2025) similarly reports that 48% of HR leaders say demand for new skills is evolving faster than existing talent structures can support, and that employees hired for “promise” rather than current proficiency are 1.9 times more likely to perform effectively.

For counseling practice, the implication is that human skills—communication, leadership, collaboration, empathy, adaptability, and critical thinking—are not auxiliary virtues but central employability assets in AI-capable workplaces.

Where Labor-Market Opportunity Is Concentrated

The U.S. Bureau of Labor Statistics (2025) Occupational Outlook Handbook identifies adjacent digital and analytical fields as continuing high-growth areas. Data scientists are projected to grow 34% from 2024 to 2034, and information security analysts 29% over the same period. By contrast, computer programmers are projected to decline 6%—a reminder that not all technology-adjacent roles share the same trajectory. Total U.S. employment is still projected to grow by 5.2 million jobs in this period.

For students, the most useful counseling guidance is often sector-first: nursing plus AI-enabled charting, business plus analytics, manufacturing plus automation literacy, cybersecurity plus governance and scripting. Pure AI specialty is one viable pathway among many; for most students, AI fluency layered onto domain expertise will prove more accessible and more durable.

Skills, Credentials, and Wages in Transition

Three intersecting trends warrant emphasis for counselors who advise students about postsecondary pathways and credentials.

First, the wage and skills evidence indicates that the labor market is increasingly rewarding demonstrable skills over credential possession alone. PwC (2025) reports both the 56% wage premium for AI-skilled workers and the faster loosening of degree requirements in AI-exposed jobs—both signals consistent with a shift toward skills-based hiring practices.

Second, employers' skills systems are not keeping up. Gartner (2025) finds that hiring for “promise” outperforms hiring for “proficiency” by a substantial margin in AI-exposed roles, suggesting that learning agility and growth mindset are becoming valued employer signals in their own right.

Third, evidence-based hiring resources continue to expand. Job-posting analyses, skills taxonomies, and credential frameworks are increasingly used to match students with opportunities. For counselors, this

trend supports advising practices that emphasize evidence of capability—projects, certificates, internships, dual-enrollment artifacts, and other tangible proof—alongside or instead of credential pursuit alone.

California's Pathway Infrastructure

For California-based counselors, the labor-market story sits alongside a strong and well-developed pathway infrastructure already serving students across the state.

The California Community Colleges system enrolls approximately 2.2 million students annually, making it the state's primary pipeline for both transfer preparation and workforce-aligned career education (California Community Colleges Chancellor's Office, n.d.-a). California's dual-enrollment system has rebounded strongly: the state reached nearly 165,000 dual-enrollment students in recent years—approximately one-third of the high school class of 2025—with community colleges providing 97% of dual enrollment statewide at little or no cost (Public Policy Institute of California, 2025b).

College and Career Access Pathways (CCAP) partnerships now account for 45% of dual-enrollment activity in California. The Los Angeles and Orange County region accounts for 29% of the state's dual enrollees, making it the most concentrated regional dual-enrollment area in California. Equity gaps remain visible in this growth: Latino and Black students remain underrepresented in dual enrollment overall, even as CCAP participation has been more racially representative than traditional dual-enrollment patterns (Public Policy Institute of California, 2025b).

Dual enrollment now produces meaningful college milestones at scale before high school graduation. In recent California data, more than 23,000 dual enrollees completed English composition, more than 18,000 completed transfer-level math, and approximately 1,950 students earned associate degrees while still in or just out of high school. The class of 2024 dual enrollees averaged 8.5 college credits earned by graduation (Public Policy Institute of California, 2025b).

Career and technical education outcomes from the California Community Colleges Chancellor's Office (2024) Strong Workforce Program legislative report indicate that 74% of Strong Workforce students secured a job in a field related to their training, 47% reached a living-wage threshold, and the median change in earnings was 30%. Notably, 69% of Strong Workforce participants were economically disadvantaged students—evidence that these pathways serve as meaningful mobility ladders, not only as career-preparation tracks for already-advantaged students.

Local pathway infrastructure further supports this work. Long Beach City College's Early College Initiatives, CCAP partnerships, and the Long Beach College Promise (which provides two years of free tuition to eligible local graduates) collectively support disproportionately impacted students through tuition-free college credit, structured supports, and 30+ Career Education programs (Long Beach City College, n.d.-d, n.d.-b, n.d.-e). LBCC enrolled 38,696 students in 2023–24, including a 10% increase in Black student enrollment in Fall 2023.

Considerations for Counseling Practice

The convergence of workforce-change evidence and California-specific pathway infrastructure supports several considerations for counseling practice. These are framed not as prescriptive directives but as evidence-aligned approaches that counselors may incorporate based on local context and student needs.

Reframe the advising question from “Which jobs are safe?” to “Which tasks, tools, and skills define this pathway now?” The available evidence consistently shows that change is task-level rather than job-level for most occupations (OECD, 2024; World Economic Forum, 2025). Asking task-level questions tends to produce more actionable advising conversations and less anxiety-driven decision-making among students.

Treat AI literacy as baseline career readiness, not a niche specialty. Workers in health, business, public service, education, the creative industries, manufacturing, and logistics increasingly need the ability to use AI tools appropriately—not only those headed into computer science or engineering (LinkedIn Economic Graph, 2025; PwC, 2025).

Keep human skills at the center of advising. Communication, leadership, collaboration, empathy, adaptability, and critical thinking are not “nice extras” but the skills most often paired with AI-capable work in employer demand data (Deloitte, 2025; OECD, 2024).

Promote sector-first pathways with AI fluency layered on top. For most students, the strongest preparation is sector-specific expertise plus AI tool fluency: nursing plus AI-enabled charting, business plus analytics, manufacturing plus automation literacy. Pure AI specialization is a viable pathway for some students, but it is not the only or the most accessible route to AI-capable employment (U.S. Bureau of Labor Statistics, 2025).

Use skills-based advising language. Help students describe what they can do with concrete evidence: prompt design, data cleaning, quality checking, documentation, customer communication, workflow improvement, and secure handling of information. This language aligns with employer movement toward skills signals (Gartner, 2025; PwC, 2025).

Direct students toward applied experiences early. Dual enrollment, CTE, job shadowing, internships, micro-internships, industry projects, and college courses that produce tangible artifacts have grown in employer value as both employers and young workers signal demand for practical preparation (Deloitte, 2025; Public Policy Institute of California, 2025b).

Maintain explicit attention to equity. California’s dual-enrollment expansion is substantial, but participation and outcomes remain uneven by race, gender, and region. Counselor outreach, multilingual communication, and intentional recruitment remain essential to ensure that pathway infrastructure benefits the students who need it most (Public Policy Institute of California, 2025a, 2025b).

Teach students to verify AI output. The strongest field evidence shows that AI can speed work substantially on appropriate tasks and degrade performance on inappropriate ones (Dell’Acqua et al.,

2025). Advising language that emphasizes “check, test, cite, and revise” supports both academic integrity and durable employability.

A useful synthesis frame for counselors is the following: AI adoption changes tasks; task change changes skills; counselors then help students convert those skills into pathway experiences and proof of capability. Each link in this chain is supported by current evidence and connects directly to existing California pathway infrastructure.

Limitations and Open Questions

Several limitations of the current evidence base warrant acknowledgment when this synthesis is applied to advising practice.

Long-run net employment effects remain uncertain. Employer surveys, exposure studies, and field experiments are well-suited to understanding adoption, tasks, and skills, but they do not yet settle the long-run net employment effect of AI on the U.S. labor market (OECD, 2024). Counselors should be cautious about making strong predictive claims about specific occupations’ futures.

Some California career-education outcomes are lagged. Strong Workforce employment and earnings metrics are retrospective and reflect students who completed pathways in prior years. They are useful for evaluating the historical performance of the pathway system, but they should not be interpreted as a real-time labor-market dashboard (California Community Colleges Chancellor’s Office, 2024).

Job-posting data are useful but not complete. Several major findings in this synthesis rely on Lightcast and similar vacancy datasets, which are strong demand signals but do not directly measure all hiring activity, all employers, or actual on-the-job AI use (OECD, 2024).

These limitations counsel humility in advising language: counselors can confidently describe current trends and pathway opportunities, but they should avoid overclaiming about specific career outcomes for individual students.

Conclusion

For K–16 counselors in California, the available evidence base supports a clear practice frame. AI is changing tasks and skills faster than it is changing employment levels for most occupations. Students benefit when counseling conversations focus on the skills and evidence required for specific pathways—including AI literacy, durable human skills, sector-specific expertise, and demonstrable capability—rather than on jobs-replacement narratives. California already maintains substantial pathway infrastructure to support this work, including dual enrollment, CCAP, College Promise programs, Strong Workforce CTE, certificates, and work-based learning. The opportunity for counselors is not to construct new systems but to connect students more intentionally to systems that already exist.

References

- Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). *Generative AI at work* (NBER Working Paper No. 31161). National Bureau of Economic Research. <https://www.nber.org/papers/w31161>
- California Career Center. (n.d.). <https://www.calcareercenter.org/>
- California Community Colleges Chancellor's Office. (n.d.-a). *State of the system*. <https://www.cccco.edu/About-Us/State-of-the-System>
- California Community Colleges Chancellor's Office. (n.d.-b). *Strong Workforce Program*. <https://www.cccco.edu/About-Us/Chancellors-Office/Divisions/Workforce-and-Economic-Development/Strong-Workforce-Program>
- California Community Colleges Chancellor's Office. (2024). *2023–24 Strong Workforce Program legislative report*. <https://www.cccco.edu/-/media/CCCCO-Website/docs/report/202324strongworkforceprogram14a11y.pdf>
- CaliforniaColleges.edu. (n.d.). <https://www.californiacolleges.edu/>
- CareerOneStop. (n.d.). *Skills tools*. U.S. Department of Labor. <https://www.careeronestop.org/Toolkit/Skills/skills.aspx>
- Deloitte. (2025). *2025 Gen Z and millennial survey*. Deloitte Insights. <https://www.deloitte.com/us/en/insights/topics/talent/2025-gen-z-millennial-survey.html>
- Dell'Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Kraymer, L., Candelon, F., & Lakhani, K. R. (2025). Navigating the jagged technological frontier: Field experimental evidence of the effects of artificial intelligence on knowledge worker productivity and quality. *Organization Science*. <https://pubsonline.informs.org/doi/10.1287/orsc.2025.21838>
- Gartner. (2025, March 11). *Gartner HR survey reveals hiring for promise instead of proficiency is more effective and efficient for closing skills gaps* [Press release]. <https://www.gartner.com/en/newsroom/press-releases/2025-03-11-closing-skills-gaps-at-scale>
- LinkedIn Economic Graph. (2025). *Work change report*. LinkedIn. <https://economicgraph.linkedin.com/content/dam/me/economicgraph/en-us/PDF/Work-Change-Report.pdf>
- Long Beach City College. (n.d.-a). *Career Center*. <https://www.lbcc.edu/career-center>
- Long Beach City College. (n.d.-b). *College & Career Access Pathways (CCAP)*. <https://www.lbcc.edu/post/college-career-access-pathways-ccap>
- Long Beach City College. (n.d.-c). *Counseling*. <https://www.lbcc.edu/counseling>
- Long Beach City College. (n.d.-d). *Early College Initiatives*. <https://www.lbcc.edu/earlycollege>

- Long Beach City College. (n.d.-e). *Long Beach College Promise*. <https://www.lbcc.edu/post/long-beach-college-promise>
- Los Angeles Regional Consortium. (n.d.-a). *Dual enrollment*. <https://losangelesrc.org/k12-strong-workforce-program/dual-enrollment/>
- Los Angeles Regional Consortium. (n.d.-b). *Regional resources*. <https://losangelesrc.org/resources/>
- McKinsey & Company. (2024). *The state of AI: How organizations are rewiring to capture value*. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-how-organizations-are-rewiring-to-capture-value>
- McKinsey & Company. (2025). *Superagency in the workplace: Empowering people to unlock AI's full potential at work*. <https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/superagency-in-the-workplace-empowering-people-to-unlock-ais-full-potential-at-work>
- Organisation for Economic Co-operation and Development. (2024). *Artificial intelligence and the changing demand for skills in the labour market*. OECD Publishing. https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/04/artificial-intelligence-and-the-changing-demand-for-skills-in-the-labour-market_861a23ea/88684e36-en.pdf
- Organisation for Economic Co-operation and Development. (2025). *Generative AI and the SME workforce*. OECD Publishing. https://www.oecd.org/en/publications/generative-ai-and-the-sme-workforce_2d08b99d-en/full-report/component-5.html
- PricewaterhouseCoopers. (2025). *The fearless future: 2025 global AI jobs barometer*. <https://www.pwc.com/gx/en/issues/artificial-intelligence/job-barometer/2025/report.pdf>
- Public Policy Institute of California. (2025a). *Dual enrollment in California*. <https://www.ppic.org/publication/dual-enrollment-in-california/>
- Public Policy Institute of California. (2025b). *Dual enrollment in California* [Fact sheet]. <https://www.ppic.org/publication/fact-sheet-dual-enrollment-in-california/>
- U.S. Bureau of Labor Statistics. (2025). *Data scientists: Occupational outlook handbook*. U.S. Department of Labor. <https://www.bls.gov/ooh/math/data-scientists.htm>
- World Economic Forum. (2025). *The future of jobs report 2025*. <https://www.weforum.org/publications/the-future-of-jobs-report-2025/digest/>

Appendix: Selected Resources for K–16 Counselors

The resources below provide direct support for counselors implementing pathway-aware advising in California, with a particular focus on the Los Angeles region and Long Beach City College. URLs are current as of publication.

Resource	What it offers	Official link
CaliforniaColleges.edu	California's official college and career planning platform for students and schools	https://www.californiacolleges.edu/
California Career Center / CareerZone	Career exploration, interest tools, and occupation information for California students	https://www.calcareercenter.org/
CareerOneStop Skills Tools	Federal tools for skill, interest, and values matching	https://www.careeronestop.org/Toolkit/Skills/skills.aspx
LBCC Early College Initiatives	Dual-enrollment bridge for middle and high school students with an equity focus	https://www.lbcc.edu/earlycollege
LBCC College & Career Access Pathways (CCAP)	Tuition-free dual-enrollment pathway designed to help students earn 12+ units by graduation	https://www.lbcc.edu/post/college-career-access-pathways-ccap
LBCC Long Beach College Promise	Two years of free tuition for eligible local graduates plus structured supports	https://www.lbcc.edu/post/long-beach-college-promise
LBCC Career Center	Career exploration, internships, résumé support, appointments, and Handshake; student appointments line: (562) 938-5195	https://www.lbcc.edu/career-center
LBCC Counseling	Academic, career, transfer, and educational planning support with appointments and chat	https://www.lbcc.edu/counseling
LARC Regional Resources	LA Career Finder, CTE Matrix, dual-enrollment and work-based-learning resources	https://losangelesrc.org/resources/
LARC Dual Enrollment Hub	Regional dual-enrollment support and pathway coordinator contacts	https://losangelesrc.org/k12-strong-workforce-program/dual-enrollment/
California Community	Statewide CTE metrics, regional	https://www.cccco.edu/About-Us/Chancellors-

Resource	What it offers	Official link
Colleges Strong Workforce Program	plans, and legislative reports	Office/Divisions/Workforce-and-Economic-Development/Strong-Workforce-Program